Tentative project title: Network-on-Chip (NoC) Topology Characterization Under Diverse Traffic Workloads

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Overview of the project topic (~5 sentences): Networks on chips (NoCs) are communication networks used to connect processor cores, caches, and other subsystems in modern multicore processors, and they are one of the fields we are highly interested in. The topology of a NoC dictates how different nodes are connected in a processor. As the number of nodes (on-chip components) increases, communication between them becomes a major factor affecting latency, throughput, and overall system performance. Different workloads can also produce varying communication patterns, so no particular topology is the best for every workload. We want to use this project to explore NoCs in detail as graphs and analyze how their structural properties interact with different traffic patterns to affect network performance.

Goal of the project (~5 sentences): For our project, we plan to explore different traffic patterns between cores and how different NoC topologies behave under those conditions. Specifically, we aim to use graph analysis to characterize the structural properties of each topology and understand why certain topologies perform better under specific traffic patterns. We will measure their average packet latency, queueing latency, average hop count, and more during this simulation. We also plan to examine NoC topologies utilized in modern CPUs and accelerators and apply our analysis to understand their design choices. Through this work, we hope to establish a relationship between communication topology affects performance and identify factors that should be considered when making NoC design decisions.

Dataset and networks (~6 sentences):

For our network and dataset, we will use the gem5 Garnet library to generate different traffic patterns. According to the documentation, garnet_synth_traffic.py allows us to configure the traffic pattern and different network topologies for each simulation. In the NoC network, nodes will represent routers or processing cores, and edges will be the communication links between them. We will study popular topologies, including Mesh, Ring, Tree, and Torus, and characterize graph properties such as degree, average shortest path length, diameter, connectivity, and betweenness centrality. As mentioned above, we will run different traffic patterns, such as uniform random, tornado, bit complement, bit reverse, neighbor, shuffle, and more, provided by Garnet, on these topologies and collect the necessary data from the output file. From this data, we will analyze the relationship between measured network performance and the structural properties used by these topologies.

Division of labor (~5 sentences):

Both members will work together to develop different topologies and their characteristics before proceeding to experimentation. We will also discuss potential hypotheses about which topology would deliver the best performance for each workload, based on the analysis. For experimentation, Steve will work on setting up the gem5 environment and writing the necessary script to run simulations. Kushal will run the simulation and collect latency and other performance metrics. Once the experiment is done, we will divide the number of topologies we tested between the two members, with each person responsible for analyzing the latency results and determining the cause. After the individual analysis, we will discuss the results and findings across all topologies. To ensure we stay on track, we will regularly meet to update on our progress and make a clear milestone deadline for ourselves, such that we are able to complete the project in a timely manner.